



Assessment of an aqueous seed extract of *Parkia clappertoniana* on reproductive performance and toxicity in rodents



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ABSTRACT

Ethnopharmacological relevance: The seeds of *Parkia clappertoniana* Keay (Family: Fabaceae) are extensively used in food in the form of a local condiment called 'Dawadawa' in Ghana and consumed by all class of people including sensitive groups such as pregnant women and children. Also, crudely pounded preparations of *P. clappertoniana* seeds are used as labor inducing agent in farm animals by local farmers across northern Ghana where nomadism is the livelihood of most indigenes. Ecologically, *P. clappertoniana* is extensively distributed across the savannah ecological zone of many African countries where just like Ghana it enjoys ethnobotanical usage. Although, many studies have investigated some aspects of the pharmacological activity of *P. clappertoniana*, none of these studies focused on the reproductive system, particularly its effects on reproductive performance and toxicity. To contribute, this study assessed the effect of aqueous seed extract of *P. clappertoniana* (PCE) on reproductive performance and toxicity in Sprague-Dawley rats and ICR mice. **Methods:** After preparation of PCE, it was then tested on rodents at different gestational and developmental windows (1–7, 8–14, and 15–term gestational days) to assess the following: mating behavior, implantation rate, maternal and developmental toxicities. Generally, animals were randomly grouped into five and treated as follows: normal saline group (5 ml/kg *po*), cytotec (misoprostol) group (200 mg/kg *po*), folic acid group (5 mg/kg *po*), and PCE groups (100, 200, and 500 mg/kg *po*), however, these groupings were varied to suit the specific requirements of some parameters. For acute toxicity, animals were orally administered PCE (3 and 5 g/kg for mice and rats respectively). **Results:** PCE-treated rats showed improved mating behavior compared to control rats. PCE improved implantation rate compared to misoprostol-treated rats. On the average, PCE-treated rats delivered termed live pups at 21 days compared to that of folic acid-treated rats at 23 days. Also, PCE-treated rats showed no observable maternal and developmental toxicities compared to folic acid and control rats. PCE (3–5 g/kg *po*) was orally tolerated in rodents. **Conclusion:** Oral administration of *Parkia clappertoniana* seed extract improves reproductive performance in rodents with no observable maternal and developmental toxicity.

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1. Introduction

Since antiquity, herbal resources from indigenous cultures continue to serve mankind in many capacities including food, disease remedy, fertility, aphrodisiac (sexual enhancement), and contraception (Rousseaux and Schachter, 2003). Currently, there is rising concerns with regards to safety and critical information

pertaining to scientific validation of herbal resources commonly used by indigenous people, yet use of herbal resources, particularly herbal medicines continue to be popular worldwide (Rousseaux and Schachter, 2003). In Ghana, a tropical West African country, many herbs continue to be used in various forms as food and remedy for humans and their farm animals (Moyo et al., 2015; Soelberg et al., 2015). Specifically, in northern Ghana herbs do not only form an integral component of common foods such as 'Tuo-zaafi' but also represent an important first line remedy for the treatment of common diseases (Moyo et al., 2015). One of the commonest plants with widespread ethnobotanical usage as food, herbal remedy as well as a veterinary medicine in northern Ghana

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is *Parkia clappertoniana*. The seeds of *P. clappertoniana* are consumed in various forms by many Ghanaians including sensitive groups (pregnant women and children). For example, a local fermented seed condiment made from *P. clappertoniana* locally called 'Dawadawa' in Ghana (Hens, 2006) is a delicacy in northern Ghana as a key ingredient for soup preparation. Also, crude preparations of the fresh seeds of *P. clappertoniana* are used to augment labor in farm animals, especially in the event of prolonged or complicated labor. The use of the seeds of *P. clappertoniana* as a veterinary medicine became apparent courtesy a survey carried out in the northern region of Ghana and this was subsequently confirmed by one Mr. Alfred Danana Samba (A retired senior veterinary officer, now resident in Wa, Upper West region of Ghana). Apart from Ghana, *P. clappertoniana* is also common in many other African countries especially those within the Savannah ecological zone, where just like Ghana, it enjoys extensive ethnobotanical usage (Boye et al., 2014). For instance, in La Cote D'ivoire, a neighboring country with Ghana, *P. clappertoniana* is called 'Soumaria' and 'Le nere' locally and in French respectively, and the seeds are used in the preparation of many kinds of food.

In view of the ethnobotanical value of *P. clappertoniana*, especially in Ghana and other African countries, many preliminary pharmacological studies have been conducted on some aspects of the pharmacology of *P. clappertoniana* seeds reflective of the ethnobotanical context in which they are used including but not limited to hepatoprotective activity (Patrick-Iwuanyanwu et al., 2010), anti-diabetic/anti-hyperlipidaemic activity (Odetola et al., 2006), anti-microbial effect (Banwo et al., 2005), anti-ulcer/anti-leprosy effects (Udobi et al., 2008), renal protection and anti-oxidant activities (Boye et al., 2014), molluscicidal activity (Lemmich et al., 1996) and nutritional properties (Hens, 2006; Ijorotimi and Keshinro, 2012).

Nonetheless, a thorough search on pubmed and other major scientific databases revealed apparent lack of published scientific information on *P. clappertoniana* seeds with respect to the reproductive system. Lack of safety data specifically on the effects of *P. clappertoniana* seeds on the reproductive system is crucial in view of its widespread usage in many African countries and also recent growing concerns on the safety of some commonly used herbal products. For instance, a recent report cited reproductive toxicity associated with some herbs used as food and remedy elsewhere (Wang et al., 2012). Relatedly, it was reported that some Chinese herbal medicines hitherto considered safe and effective (Marcus and Snodgrass, 2005; Ong et al., 2005; Westfall, 2001) produced serious congenital malformations in pregnant mice (Wang et al., 2012). We strongly suspect that with the long uneventful use of *P. clappertoniana* seeds in Ghana as food and remedy, it may present no maternal and developmental toxicities but rather improve reproductive performance. To test our suspicion, and also to provide rationale for future study on *P. clappertoniana* seeds, the present study assessed the effect of aqueous seed extract of *P. clappertoniana* (PCE) on reproductive performance, maternal and developmental toxicity in rodents.

2. Materials and methods

2.1. Drugs and chemicals

Drugs and chemicals secured for the study included Misoprostol (Euro Biotech, Chandigarh, Chandigarh-160002, India), Folic acid (Alpha Chemika, Mumbai, India) pentobarbital sodium (BDH Industries Limited), normal saline (Lupin Pharma Limited, Mumbai, India). All experiments were conducted at the laboratories of the Department of Pharmacology, Center for Scientific Research into Plant Medicine (CSRPM), Mampong-Akuapim in the

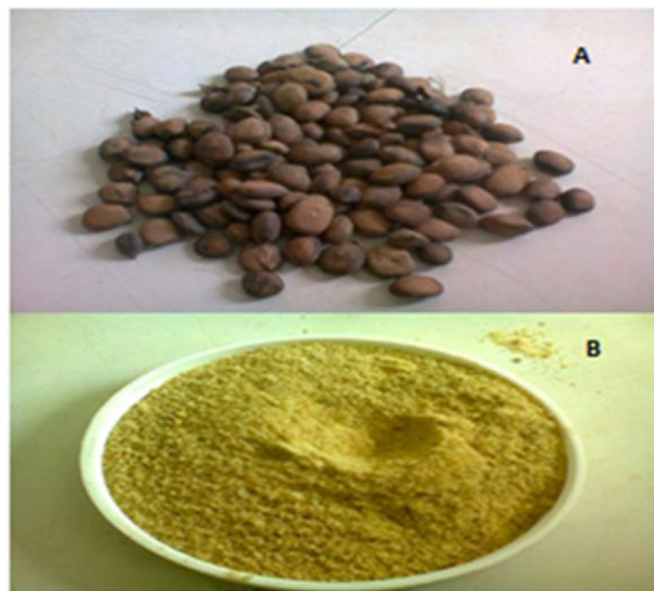


Fig. 1. (A) Seeds of *Parkia clappertoniana* showing the dark-brownish husk and (B) yellow powdered seeds of *Parkia clappertoniana* after boiling-aided dehusking and drying. Subsequently, aqueous extract of the yellow powder was made and used for the study. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Eastern region of Ghana.

2.2. Collection, identification and authentication of *Parkia clappertoniana* seeds

The seeds of *Parkia clappertoniana* were obtained from local markets in Wa, the capital of the Upper West Region of Ghana. Identification and authentication of the seeds were done by Prof. Henry Akotoye, a plant taxonomist at the Department of Botany, School of Biological Sciences, University of Cape Coast, Ghana, where a voucher specimen (UCC/SBS/P133) was deposited at the herbarium unit for reference as previously reported (Boye et al., 2014).

2.3. Preparation of aqueous Seed extract of *Parkia clappertoniana* (PCE)

The aqueous seed extract of *Parkia clappertoniana* was prepared by following the method previously described (Boye et al., 2014). Concisely, a 1 kg amount of *P. clappertoniana* seeds (Fig. 1) were dehusked allowed to ferment, sun-dried, pulverized, steam-boiled and allowed to cool down. The cold infusion was filtered (Whatman/Schleicher & Schuell, GmbH, Austria), and filtrate frozen in a pre-freezer (S-IF, Rikakikai Co. Ltd Tokyo, Japan) and dried (Heto Power Dry LL 300 freeze-dryer, Thermo Electron Corporation, USA) for a week to produce PCE. The percentage yield of PCE was 31.5%. The powdered PCE was stored in a refrigerator at -80°C until use.

2.4. Experimental animals

Healthy and matured eight weeks old Sprague-Dawley rats of either sex weighing (250–300 g) and imprinting controlled region (ICR) mice weighing (30–45 g) were used in this study. All experimental animals were obtained from the Animal Breeding Unit of CSRPM. Both rats and mice were kept in aluminium cages (60 × 30 × 25 cm), and fed on pelleted feed (Agricare Ltd, Kumasi, Ghana) and allowed free access to clean water, however, this was varied to suit the specific requirements of some experiments. All

animals were kept under ambient conditions of light/dark cycle, humidity and room temperature throughout the study. All the requirements with regards to humane handling and use of animals in experiments as enshrined in the “Principles of laboratory animal care” (NIH publication No. 85-23, revised 1985) as well as institutional and specific national laws were strictly adhered to.

2.5. Assessment of the effect of PCE on mating behavior of rats

Both female (5/treatment group) and male (3/treatment group) rats in separate groups were initially treated daily with normal saline (5 mg/kg body weight *po*) and PCE (100 and 500 mg/kg body weight *po*) for two weeks. Subsequently, female and male rats were sub-grouped together for cohabitation for one more week at a ratio of 1: 3 (Male: female) followed by daily drug treatment for seven days. A rat in each cohabitation group was uniquely labeled to permit individual observation. Rats in their cages were placed in a testing chamber with a mirror placed at 45° beneath the testing chamber which made possible an unobstructed view of the rats in the testing chamber. The mating behaviors of the rats were captured (60 min in the case of each rat) by using a camcorder (Evario™ model GZ-MG1300, JVC, Tokyo, Japan) which was positioned in front of the mirror. The mating behaviors were scored (60 min) for each rat from three separate captured videos. Scoring of mating behaviors was done by three independent persons.

2.6. Assessment of the effects of PCE and cytotec on implantation in pregnant rats

Female rats (5–10/treatment group) were initially treated for a period of one week as follows: group one: normal saline (5 ml/kg *po*), group two: cytotec (200 mg/kg body weight *po*), and groups three, four and five: PCE (100, 200, and 500 mg/kg body weight *po*). Subsequently, the female rats of comparable body weights without confirmed stage of estrous cycle were sub-grouped with confirmed fertile male rats for one week cohabitation at a ratio of 1: 3 (Male: female) and drug treatments continued. Confirmation of pregnancy in rats post-coitus was established by observing a physical presence of a vaginal plug as previously described (Lee and DeMayo, 2004). On day 6 of cohabitation or post-coitus, each of the female rats in each group was sacrificed, uterus isolated, wet weight of uteri measured and the number of implants (Determined by counting the number of implantation sites or invaginations/dam for each treatment group) in each uterus determined for each treatment group. This whole experiment was repeated, but this time the number of implants was determined on day 15 of cohabitation or post-coitus. Briefly, each female rat was sacrificed by deep anesthesia using pentobarbital (60 mg/kg *ip*) and the abdomen surgically opened to expose the Y-shaped uterus. The two horns of each uterus were separated and freed from fat and connective tissues. Each uterine horn was dorso-longitudinally cut open so that it was now a sheet of muscle instead of a narrow tube. The pregnancy status and the number of implantation spots were determined for female rats in each treatment group.

2.7. Assessment of the effects of PCE and folic acid on gestation period, litter size, litter weight and teratogenicity

Confirmed pregnant rats (Detection of copulatory plug or semen at the mouth of the female copulatory organ) (Lee and DeMayo, 2004) of comparable body weights were randomly assigned to one of five groups (5–10/treatment group) and treated as follows: group one (Control: in addition to daily access to water and chow, received normal saline: 5 ml/kg bodyweight *po*), group two

(Received folic acid: 5 mg/kg bodyweight *po*) and groups three, four and five (Received PCE: 100, 200 and 500 mg/kg bodyweight *po*) for 18 days. The time for daily treatment of drugs was uniform for the entire treatment period. After grouping, rats were assigned to individual aluminium cages. Pregnant rats were monitored daily for clinical signs of maternal toxicity (Sedation, hypothermia, piloerection or aversion to handling and drug treatment). Additionally, the bodyweight of each pregnant rat in each group was measured on 1, 7, 14 and 20 gestational days by using a weighing balance (ISHIDA number 15,613, Okazaki Sanyo Company limited, Japan). From day 19 onwards pregnant rats were monitored hourly for parturition up to gestational day 23. On the day of delivery, the following parameters: number of pups delivered, number of live pups, day zero pup bodyweight, number of dead pups or pups with congenital malformations, if any were all determined for each dam in each treatment group.

2.8. Acute toxicity studies

Acute toxicity study was carried out by single oral dose administration to either sex of ICR mice (3 g/kg *po*; N=7 in each sex group) and rats (5 g/kg *po*; N=5 in each sex group) according to the Organization of Economic Co-operation and Development (OECD) guidelines for testing chemicals (OECD, “Test guideline 425, 2001) and World Health Organization guidelines (WHO, General Guidelines for Methodologies on Research and Evaluation of Traditional Medicine, 2000). Briefly, after a 24 h starvation, PCE (3 and 5 g/kg *po* for mice and rats respectively) was administered to animals in each of the four groups. Each animal was then observed 3 hourly for 24 h for any mortality or clinical symptoms/signs of acute toxicity such as loss of consciousness, salivation, muscle tremor, changes in gait, ataxia and loss of reflexes.

2.9. Statistical analysis

Data presented are mean $\pm$ SEM. The level of significance was established using one-way or two-way analysis of variance (AN-OVA) followed by Bonferroni's multiple comparison test. $P \leq 0.05$ was considered statistically significant.

2.10. Theory/Calculations

A plant may contain vitamins essential for good health; nitrogenous organic compounds (alkaloids) that can stimulate almost every system of the human body including but not limited to the reproductive system, vascular and central nervous systems; antibiotics that attack microorganisms; essential oils, heterosides (glucides or sugars), acids, or minerals that interact chemically to affect certain body organs selectively. Nonetheless, the bane of herbal medicine usage is that it has usually been practiced without the appropriate scientific framework to ensure that cause and effect relationships are properly established and documented to constitute a system of verification or assessment of the efficacy of a particular herbal remedy. Thus, whereas, the local herbal medicine practitioner or user is quick to profess the efficacy of his remedy, he stops short of providing a sound scientific basis and explanation for his remedy and more often than not relies on the long, continuous uneventful use of the remedy for treatment. As part of the efforts to promote the use of herbal medicines either as an alternative or adjunct to conventional medical care or to bridge the gap between conventional and herbal medicines, it is important for natural product scientists, ethnopharmacologists, pharmacognosists, ethnobotanists, and pharmacologists just to mention but a few, collaborate at multi-level to carry out rigorous scientific investigations into herbal medicines. These scientific investigations could give birth to novel plant-derived candidate

Table 1
Effect of PCE on mating behavior in female and male rats.

Sex	Mating behavior	Normal saline	PCE (mg/kg)	
			100	200
Female	Wiggling of ears	+	+	+
	Lordosis	–	–	–
	Hopping and darting around males	+	+	++
Male	Sniffing the mouth and anogenital region of females	+	+	++
	Penile erection	–	–	–
	Mounting on females	–	–	–

+ Mating behavior was scored 50% by 3 independent assessors; ++ mating behavior was scored > 75% by 3 independent assessors. The frequency of each mating behavior was scored within 10 min time blocks for an hour each from videos captured at 3 different days.

drugs and also afford accurate prediction of the pharmacological activities of herbal medicines in the light of the ethnobotanical context in which they are used.

3. Results

3.1. Assessment of the effect of PCE on mating behavior in both female and male rats

There was no difference in mating behavior exhibited by rats in both PCE (100 mg/kg) and normal saline treatment groups. However, female rats in PCE (200 mg/kg) treatment group demonstrated enhanced wiggling of ears, hopping and darting around males, while male rats sniffed the mouth and anogenital region of females compared to rats in normal saline (Control) treatment group. Both PCE (100 and 200 mg/kg) and normal saline treatments could not induce lordosis, penile erection and female mounting in the male rats (Table 1).

3.2. Assessment of the effects of PCE and Misoprostol (Cytotec) treatments on pregnancy and implantation

At day 6 post-coitus, the number of pregnant rats, mean weight of uterus and the number of implants of misoprostol treated rats (Model) were lower compared to control group. However, there was a dose-dependent increase in the number of pregnant rats, mean uterus weight and number of implants in PCE treated rats compared to misoprostol treated and control rats. This trend was not different at day 15 post-coitus, except that PCE (100 mg/kg) produced a significant increase in the number of implants than

that of PCE (200 mg/kg) (Table 2).

3.3. Effects of PCE and folic acid treatments on bodyweight changes in pregnant rats

There were no mortality and clinical signs of maternal toxicity across all groups during treatment. Compared with normal saline treated rats, folic acid treatment significantly increased bodyweight of pregnant rats for the first (1–7 days) and last (14–20 days) 7 days of gestation. Within the same period, PCE treated rats produced a comparably higher increases in bodyweight changes than folic acid and normal saline treated rats. From 7 to 14 days of gestation, the bodyweight changes between folic acid and normal saline treated rats were not significantly different. For the same period, the difference in bodyweight changes of PCE (200 mg/kg) treated rats were not significantly different from that of folic acid and normal saline treated rats, but that of PCE (100 and 500 mg/kg) were significant compared to folic acid and normal saline treatment groups (Tables 3 and 4).

3.4. Effects of PCE and folic acid treatments on sex ratio of pups and length of gestation

Sex ratio (Male fraction) of pups from rats treated with folic acid and normal saline was not significantly different as most were males. Except PCE (500 mg/kg) which produced sex ratio (Male fraction) comparable to folic acid and normal saline groups, PCE (100 and 200 mg/kg) produced sex ratios (Male fraction) which were significantly different from that of folic acid and normal saline groups as most of the pups were females.

Folic acid treatment produced a significant ($P \leq 0.05$) increase in the length of gestation period in pregnant rats compared to that of normal saline treated pregnant rats. However, PCE treated rats delivered at a shorter length of gestation compared to that of folic acid treated rats (Table 5).

3.5. Effects of folic acid and PCE treatments on litter size and litter weight

No dead pup was recorded for any of the treatment groups. Also, pups from all the treatment groups showed no any observable physical congenital malformations. No pup died post-natal day 5 across all groups and pathological examination of pups showed no abnormality. The difference in average litter size of folic acid and normal saline treated rats was not significantly different. Also, the mean litter size of PCE (200 mg/kg) treated rats was comparable to that of folic acid and normal saline. Although, the mean litter size of PCE (100 and 500 mg/kg) was numerically different from that of folic acid and normal saline treatment groups, it was statistically insignificant (Table 6).

Table 2
Effect of PCE on implantation rate and uterine weight.

Treatment groups		Gestational day 6		Gestational day 15	
		Wet Weight of uterus (g)	Number of Implants	Wet weight of uterus (g)	Number of implants
	^a Number of pregnant rats				
Normal saline (5 ml/kg)	5.5	1.50 ± 1.13	4.60 ± 6.31	2.91 ± 2.90	7.00 ± 6.55
Cytotec (200 mg/kg)	2.3	0.86 ± 0.28 ^{ns}	0.00 ± 0.00	1.91 ± 2.69 ^{ns}	2.20 ± 3.90 ^{ns}
PCE (100 mg/kg)	5.5	0.94 ± 0.19 ^{ns}	1.03 ± 0.00 [#]	5.26 ± 0.85 [#]	10.80 ± 6.26 [#]
PCE (200 mg/kg)	5.5	1.14 ± 0.29 ^{ns}	1.60 ± 2.60 [#]	5.85 ± 2.53 [#]	9.00 ± 5.39 [#]
PCE (500 mg/kg)	5.5	3.65 ± 3.30 [#]	8.40 ± 7.77 [#]	6.91 ± 2.06 [#]	11.0 ± 0.71 ^{##}

Values are mean ± S.E.M (n=5). The level of significance was established using one-way analysis of variance (ANOVA) followed by Bonferroni's Multiple comparison test.

[#] $P \leq 0.05$.

^{##} $P \leq 0.01$.

^a Pregnant rats for day 6 and 15 post-coitus respectively.

^{ns} No statistical significance between comparisons, $P \geq 0.05$.

Table 3

Folic acid and PCE treatments and rat maternal bodyweight (g) measured at different gestational days.

Gestational day	Normal saline (5 ml/kg)	Folic acid (5 mg/kg)	PCE (mg/kg)		
			100	200	500
1	201.6 ± 7.6	217.6 ± 4.9*	209.4 ± 7.9 [#]	203.8 ± 12.5	206.2 ± 6.0
7	201.4 ± 10.9	220.8 ± 7.1**	215.0 ± 8.3 [#]	212.2 ± 14.0 [#]	212.0 ± 7.7 [#]
14	223.4 ± 8.9	239.2 ± 6.3*	237.2 ± 10.3 [#]	231.0 ± 10.1 [#]	239.6 ± 10.4
20	225.2 ± 9.3	247.8 ± 4.9**	249.0 ± 10.9 [#]	254.2 ± 5.3 ^{#¥}	257.2 ± 6.9 ^{#¥}

Each value is the mean ± S.E.M (n=5). $P \leq 0.05$ was considered significant in all analysis. The level of significance was established using two – way analysis of variance (ANOVA) followed by Bonferroni's Multiple comparison test.

* $P \leq 0.05$: folic acid vs normal saline.

[#] $P \leq 0.05$: PCE vs normal saline.

[¥] $P \leq 0.05$: PCE vs folic acid.

** $P \leq 0.01$.

Table 4

Rat maternal bodyweight gain (g) at different gestational windows after folic acid and PCE treatments.

Gestational window ^a	Normal saline (5 ml/kg)	Folic acid (5 mg/kg)	PCE (mg/kg)		
			100	200	500
1–7	0.2 ± –3.3	3.2 ± –2.2*	5.6 ± 0.4 ^{#¥}	8.4 ± 1.5 ^{#¥}	5.8 ± 1.7 ^{#¥}
7–14	22.0 ± 2.0	18.4 ± 0.8 ^{ns}	22.2 ± 0.8 ^{ns}	18.8 ± 3.9 ^{ns}	27.6 ± 2.7 ^{#¥}
14–20	1.8 ± –0.4	8.6 ± 1.4**	11.8 ± 0.6 ^{#¥}	14.2 ± 4.8 ^{#¥}	17.6 ± 3.5 ^{#¥}

Each value is the mean ± S.E.M (n=5). $P \leq 0.05$ was considered significant in all analysis. The level of significance was established using two – way analysis of variance (ANOVA) followed by Bonferroni's Multiple comparison test.

* $P \leq 0.05$: folic acid vs normal saline.

[#] $P \leq 0.05$: PCE vs normal saline.

[¥] $P \leq 0.05$: PCE vs folic acid.

^a Post-fertilization developmental stages.

^{ns} Not significant.

Table 5

Effect of PCE on litter size and litter weight.

Treatment groups	Litter size	Litter weight (g)	Post-natal day 5 ^a
Normal saline (5 ml/kg)	9.4 ± 0.836	5.9 ± 0.230	9.6 ± 0.011
Folic acid (5 mg/kg)	9.2 ± 0.961 ^a	6.4 ± 0.40*	9.4 ± 1.032
PCE (100 mg/kg)	10.6 ± 0.908 ^{ns}	6.1 ± 0.417 ^{ns}	10.8 ± 0.009
PCE (200 mg/kg)	9.5 ± 0.836 ^{ns}	5.9 ± 0.253 ^{ns}	9.9 ± 0.113
PCE (500 mg/kg)	10.2 ± 1.474 ^{ns}	6.3 ± 0.378 ^{ns}	10.6 ± 1.023

Each values is the mean ± S.E.M (n=5). The level of significance was established using one-way analysis of variance (ANOVA) followed by Bonferroni's Multiple comparison test. $P \leq 0.01$ was considered significant.

* $P \leq 0.01$: folic acid vs normal saline.

^a Litter size post-natal day 5.

^{ns} No statistical significance between comparisons, $P \geq 0.05$.

Table 6

Effect of folic acid and PCE treatments on sex of pups and length of full term gestation period.

Treatment groups	Sex ratio (Male: Female)	Gestational period (Days)
Normal saline (5 ml/kg)	1.21 ± 0.234	21.6 ± 0.274
Folic acid (5 mg/kg)	1.13 ± 0.277 ^{ns}	23.6 ± 0.447**
PCE (100 mg/kg)	0.93 ± 0.210 ^a	21.4 ± 0.274 [#]
PCE (200 mg/kg)	0.88 ± 0.103 ^a	21.4 ± 0.274 [#]
PCE (500 mg/kg)	1.57 ± 0.229 ^{ns}	21.2 ± 0.224 [#]

Each values is the mean ± S.E.M (n=5). The level of significance was established using one-way analysis of variance (ANOVA) followed by Bonferroni's Multiple comparison test. $P \leq 0.01$ was considered significant.

** $P \leq 0.01$: folic acid vs normal saline.

[#] $P \leq 0.01$: PCE treatments vs folic acid

^a $P \leq 0.01$: PCE vs normal saline.

^{ns} Not significant.

Folic acid treated rats produced a significant ($P \leq 0.05$) increase in the mean litter weight compared to normal saline treated rats. The mean litter weight of PCE (100 and 500 mg/kg) treated rats was comparable to that of folic acid treatment group but significantly different from that of normal saline treatment group. PCE (200 mg/kg) treated rats produced mean litter weight comparable to normal saline treatment group but significantly different from that of folic acid treatment group (Table 6).

3.6. Assessment of acute toxicity of PCE in rodents

Oral exposure of ICR mice and rats to PCE (3 and 5 g/kg *po*) respectively produced no mortality over a 24 hour period of observation and there were no observable clinical signs/symptoms of acute toxicity. Also, no observable adverse effects were detected after 72 h monitoring.

4. Discussion

This study assessed the effect of PCE on the reproductive performance and toxicity in rodents. Oral PCE treatment improved reproductive performance and produced no observable maternal and developmental toxicity compared to reference drugs (folic acid and misoprostol), and these observations are important, especially at a time when many commonly used herbal resources are identified with a spectrum of adverse events (Marcus and Snodgrass, 2005; Ong et al., 2005; Rousseaux and Schachter, 2003; Westfall, 2001).

Mating behavior displayed by sexually matured animals is the physical manifestation of the actions of sex hormones at the cellular level. From the present results, PCE treated rats of either sex

displayed enhanced mating behaviors compared to that of control rats (Table 1) and these effects were not different from that of other herbs tested on reproductive performance (Cicero et al., 2002; Zheng et al., 2000). Inferentially, PCE may modulate sex hormones, specifically those involved in mating behavior in view of the fact that hormonal regulation underpins sexual behavior. In viviparous mammals, implantation embodies intimate physical contact between the conceptus and the maternal blood supply during early development (Lee and DeMayo, 2004), and it depends on implantation-competent blastocyst and a receptive uterus (Wang et al., 2006), as well as proper regulation of uterine-receptivity hormones (Giudice, 1999; Hamatani et al., 2004; Sun et al., 2009). Misoprostol (a prostaglandin analogue), which is conventionally used for safe abortion and labor augmentation (Bracken et al., 2014; Clouqueur et al., 2014; Vogel et al., 2013) produced 100% implantation failure in pregnant rats at 6 and 15 days post-pregnancy. In rodents, implantation takes place on the 5th day post-fertilization (Singh and Kamboj, 1992), however, drug-induced interference with uterine microenvironment may render the uterus non-receptive to implantation competent blastocyst leading to implantation failure (Emiliani et al., 2005). Also, drug-induced myometrial hypermotility may lead to hastened arrival of blastocyst to an apparently a non-receptive uterus leading to implantation failure (Cummings and Perreault, 1990). Misoprostol may have blocked implantation via increased fetal resorption by disrupting hormonal and physiological processes of implantation, and this was confirmed at day 15 post-pregnancy where there was high embryo resorption in misoprostol-treated rats compared to normal saline treated rats (100% implantation rate). Unlike misoprostol-treated rats, PCE-treated rats produced a dose-dependent pro-implantation effect, especially at the highest dose (Table 2). The pro-implantation effects of PCE contrasts with that of other herbs used for improving reproductive performance in other cultures including *Lepidium meyenii* (Ruiz-Luna et al., 2005) which reportedly produced no pro-implantation effects in mice (Oshima et al., 2003; Ruiz-Luna et al., 2005) and many other medicinal herbs shown to possess abortifacient (Conway and Slocumb, 1979; Madari and Jacobs, 2004; Noumi and Tchakonang, 2001) and contraceptive/anti-fertility effects such as *Michelia champaca* (Taprial et al., 2013) *Derris brevipes* (Badami et al., 2003), *Ailanthus excelsa* (Ravichandran et al., 2007), *Hibiscus rosa Senensis* (Vasudeva and Sharma, 2008), *Archyranthes aspera* roots (Vasudeva and Sharma, 2006), and others (Kumar et al., 2012).

The pro-implantation effect of PCE (Table 2) was supported by results on termed delivery of live pups (Tables 3, 4, and 5) and this was comparable to that of folic acid-treated rats. Also, PCE treatment favored delivery of female pups at lower and middle doses but favored male pups at the highest dose (Table 6). The use of folic acid before or during pregnancy is mandatory in most countries and Ghana is no exception. Folic acid intake during pregnancy prevents neural tube defects in neonates (Faria et al., 2015; Suren et al., 2013; Veeranki et al., 2015). Of Note, no neural tube defects were observed in pups from rats treated with PCE or folic acid. However, there was a difference between average gestation in PCE and folic acid treated rats. PCE-treated rats delivered on day 21 of gestation, while folic acid-treated rats delivered on day 23 of gestation. Although these two average delivery days fall within the normal range for rats (21–24), this apparent difference could not be explained by our present results, however, speculatively it could be due to drug-induced delay in implantation (Lee and DeMayo, 2004).

Acute toxicity symptoms and signs commonly associated with oral administration of high dose of a drug were not observed in PCE (3–5 g/kg po)-treated rodents over a 72 h monitoring, indicating that PCE is orally safe and well tolerated in rodents. The pharmacological-behavioral effects of PCE in this study may be attributed

to its phytochemical constituents as is mostly the case with herbs of medicinal value (Majid et al., 2015). Coordinated uterine contraction is needed for safe spontaneous labor in humans and other viviparous mammals and it is a function of contractile proteins in the uterine musculature including CAV-1, GJAI, and PTGS2 (Elmes et al., 2015). PCE and folic acid treatments produced termed delivery of live pups (Table 6), indicating that perhaps PCE and folic acid induced normal uterine contractions by stimulating the uterine musculature in a manner that led to safe delivery of live pups. But this study was limited by our inability to establish the effect of PCE on uterine contractility and the possible regulation of the uterine contractile proteins (CAV-1, GJAI, and PTGS2) as well as the site of action. Nonetheless, this study may be useful as a basis for further studies and also may allay fears and promote rationale and evidence-based usage of *P. clappertoniana* seeds in Ghana, given that many herbal resources used just like *P. clappertoniana* have been cited in connection with a spectrum of adverse effects (Marcus and Snodgrass, 2005; Ong et al., 2005; Rousseaux and Schachter, 2003; Westfall, 2001).

5. Conclusion

Parkia clappertoniana seed extract improves reproductive performance in rodents and presents no observable maternal and developmental toxicity.

Conflict of interest

None declared.

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